Student Performance Analytics System

Project Documentation

Submitted by: Amrutha M

Project Title: Student Performance Analytics System

Technologies Used: Python, Pandas, NumPy, Matplotlib

Dataset Size: 20 student records

Project Objective

The main objective of this project is to develop and build a simple and efficient system used for analyzing the student's academic performance using python.

The project aims to:

- Organize student information using a Pandas DataFrame.
- Calculate total marks and average marks for each student.
- Assign grades based on predefined grading criteria.
- Identify students who pass or fail.
- Calculate overall class performance statistics.
- Analyze performance across different subjects.
- Identify the top-performing students.
- Represent important findings using graphs and charts.

The system uses Python, Pandas, and NumPy to calculate marks, averages, grades, and pass/fail results. It also performs subject-wise performance analysis and identifies top-performing students. Matplotlib is used to visualize the analysis through charts.

Technologies Used

1) **Python:** Python is the programming language used in this project. It manages the program logic, conditional statements, loops, functions and calculations.

2) **Pandas:** Pandas is used to read the CSV file, organize the data into data frames, create calculated columns, sort students, and analyze student and department performance.
3) **NumPy:** NumPy is used for numerical calculations, including average marks, lowest marks, highest marks and other statistical operations.
4) **Matplotlib:** Matplotlib is used to create data visualization of the analysis including subject-wise average marks, pass/fail distribution and top-performing students.
5) **Microsoft Excel:** Microsoft Excel is used to prepare and organize the initial student dataset before saving it in CSV format.
6) **Google Colab:** Colab is used to run the python code and libraries of this project/system.
7) **GitHub:** GitHub is used to store and share the project's source code, dataset, documentation and the README file.

Dataset Description

The dataset contains 20 student records. Each record includes academic and attendance information. The dataset contains the following columns:

Column	Description
Student ID	Unique identifier for each student.
Name	Student's name.
Department	Student's branch.
Mathematics	Marks obtained in the subject 'Mathematics'.
Python Programming	Marks obtained in the subject 'Python Programming'.
Engineering Drawing	Marks obtained in the subject 'Engineering Drawing'.
Attendance Percentage	Attendance percentage of the student.

Implementation

The following are the steps for implementing this project:

1) Loading and inspecting the dataset: The CSV file is loaded into the panda's data frame and the output displayed is the student's records and examines the column names, dimensions and data types.

2) Calculating the total Marks and average marks of each student: The system calculates the total marks obtained by each student across the three subjects so the total marks are calculated using the formula: Total marks = mathematics + Python programming + engineering drawing Then the program also calculates the average marks of each student So the formula used is: average marks = total marks of each student / number of subjects.

3) **Grade assignment:** The program assigns the grades according to the student's average marks. The following are the average marks and grades which are assigned to each student according to their performance:

Average Marks	Grade
90-100	A+
80-89.99	A
70-79.99	B+
60-69.99	B
50-59.99	C
Below 50	F

A Python function is used to provide grades for each student's average marks.

4) **Pass/fail classification:** The system determines the result of each student based on their subjects' marks: If a student obtains at least 40 marks in every subject then he/she is pass. Otherwise, if they score below 40 anyone of the subject also, they are considered as failed.

5) **Analysis of overall performance**: The system calculates and displays:

- The total number of students present.
- Number of students who passed.
- Number of students who failed.
- Overall class Average Marks.
- Highest student's average.
- Lowest student's average.

6) **Data Visualization:** Matplotlib, a Python library is used to present the analysis graphically or using charts.

- **Subject wise average marks:** A bar chart comparing average marks across the three subjects.
- **Student pass/fail distribution:** A pie chart showing the number of students who passed and failed.
- **Top five performers:** A bar chart comparing the average marks of the highest-performing students.

7) **Identifying Top Performers**

- The system sorts students according to their average marks in descending order and selects the top five performers.
- Their student IDs, names, departments, average marks, and grades are displayed.

Key Features

Main features of the Student Performance Analytics System include:

- Loading all the student's data (CSV file) using Pandas.
- Calculation of total and average marks of each student.
- Grades implemented automatically.
- Classification of results into either pass or fail.
- Derivation of general performance report of all the students.
- Analysis of the subject data statistically.
- Selection of top-performing students.
- Data Visualization using Matplotlib.

Output / Screenshots

The program is designed to generate the following outputs:

1) **Dataset loaded in Python**

```
    Student ID        Name  Department  Mathematics   Python programming  \
0         1001     Ritvik          CSE           56                   80
1         1002   Raghunath         ISE           67                   70
2         1003      Priya          ECE           79                   78
3         1004      Meera           CE           84                   86
4         1005     Manisha          ME           67                   85
5         1006     Madhavi         ISE           57                   87
6         1007       Leela         ISE           24                   76
7         1008      Anjali         CSE           87                   48
8         1009   Aishwarya         ECE           66                   76
9         1010      Aaryav          CE           73                   59
10        1011      Sharan          ME           95                   35
11        1012     Anuradha        CSE           64                   32
12        1013      Mahesh          ME           47                   81
13        1014       Charu         ISE           66                   83
14        1015      Prithvi         ECE           72                   46
15        1016       Sashi          ME           78                   65
16        1017       Asha           CE           77                   99
17        1018     Bhavana         ISE           93                   49
18        1019     Sinchana        CSE           89                   83
19        1020      Gaurav         ECE           88                   41

    Engineering Drawing  Attendance Percentage
0                    90                    84
1                    38                    83
2                    90                    92
3                    78                    72
4                    53                    75
5                    86                    94
6                    36                    90
7                    83                    79
8                    30                    67
9                    82                    86
10                   54                    67
11                   74                    92
12                   81                    60
13                   84                    88
14                   48                    65
15                   76                    77
16                   53                    70
17                   86                    80
18                   79                   100
19                   59                    75
```

2) First Five Records

```
-------------------------First Five Records-------------------------------
   Student ID        Name  Department  Mathematics   Python programming  \
0        1001     Ritvik          CSE           56                   80
1        1002   Raghunath         ISE           67                   70
2        1003      Priya          ECE           79                   78
3        1004      Meera           CE           84                   86
4        1005     Manisha          ME           67                   85

   Engineering Drawing  Attendance Percentage
0                   90                    84
1                   38                    83
2                   90                    92
3                   78                    72
4                   53                    75
```

3) Dataset Information

```
...
    ------------------------Dataset Information-------------------------------
    <class 'pandas.core.frame.DataFrame'>
    RangeIndex: 20 entries, 0 to 19
    Data columns (total 7 columns):
     #   Column                 Non-Null Count  Dtype
    ---  ------                 --------------  -----
     0   Student ID             20 non-null     int64
     1   Name                   20 non-null     object
     2   Department             20 non-null     object
     3   Mathematics            20 non-null     int64
     4   Python programming     20 non-null     int64
     5   Engineering Drawing    20 non-null     int64
     6   Attendance Percentage  20 non-null     int64
    dtypes: int64(5), object(2)
    memory usage: 1.2+ KB
    None
```

4) Total Marks and Average Marks of each student

```
    0     226
    1     175
    2     247
    3     248
    4     205
    5     230
    6     136
    7     218
    8     172
    9     214
    10    184
    11    170
    12    209
    13    233
    14    166
    15    219
    16    229
    17    228
    18    251
    19    188
    Name: Total Marks, dtype: int64
    0     75.33
    1     58.33
    2     82.33
    3     82.67
    4     68.33
    5     76.67
    6     45.33
    7     72.67
    8     57.33
    9     71.33
    10    61.33
    11    56.67
    12    69.67
    13    77.67
    14    55.33
    15    73.00
    16    76.33
    17    76.00
    18    83.67
    19    62.67
    Name: Average Marks, dtype: float64
```

5) Grades Calculation

```
0     B+
1      C
2      A
3      A
4      B
5     B+
6      F
7     B+
8      C
9     B+
10     B
11     C
12     B
13    B+
14     C
15    B+
16    B+
17    B+
18     A
19     B
Name: Grade, dtype: object
```

6) Pass/Fail Distribution

```
0     Pass
1     Fail
2     Pass
3     Pass
4     Pass
5     Pass
6     Fail
7     Pass
8     Fail
9     Pass
10    Fail
11    Fail
12    Pass
13    Pass
14    Pass
15    Pass
16    Pass
17    Pass
18    Pass
19    Pass
Name: Result, dtype: object
```

7) Student Performance

```
...
    ------------------Student Performance------------------------------
     Student ID       Name  Total Marks  Average Marks Grade Result
           1001   Ritvik            226          75.33    B+   Pass
           1002 Raghunath           175          58.33     C   Fail
           1003     Priya           247          82.33     A   Pass
           1004    Meera            248          82.67     A   Pass
           1005   Manisha           205          68.33     B   Pass
           1006   Madhavi           230          76.67    B+   Pass
           1007     Leela           136          45.33     F   Fail
           1008     Anjali          218          72.67    B+   Pass
           1009 Aishwarya           172          57.33     C   Fail
           1010     Aaryav          214          71.33    B+   Pass
           1011     Sharan          184          61.33     B   Fail
           1012   Anuradha          170          56.67     C   Fail
           1013    Mahesh           209          69.67     B   Pass
           1014      Charu          233          77.67    B+   Pass
           1015    Prithvi          166          55.33     C   Pass
           1016      Sashi          219          73.00    B+   Pass
           1017      Asha           229          76.33    B+   Pass
           1018   Bhavana           228          76.00    B+   Pass
           1019   Sinchana          251          83.67     A   Pass
           1020     Gaurav          188          62.67     B   Pass
```

8) Overall Performance

```
  ------------------Overall Performance-------------------------
  Total Number of Students: 20
  Number of Students Passed: 15
  Number of Students Failed: 5
  Class Average Marks: 69.13
  Highest Average Marks: 83.67
  Lowest Average Marks 45.33
```

9) Subject Wise Performance Analysis-

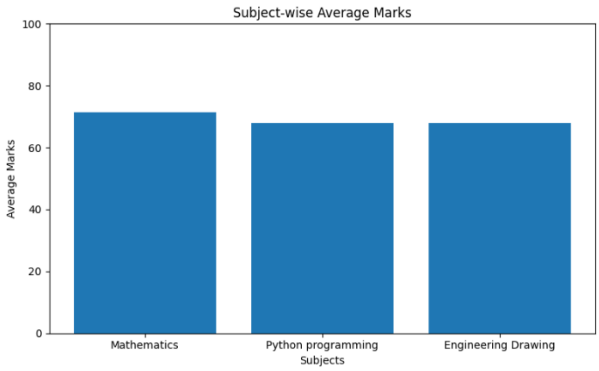
```
...
    -----------------Subject Wise Performance Analysis----------------------

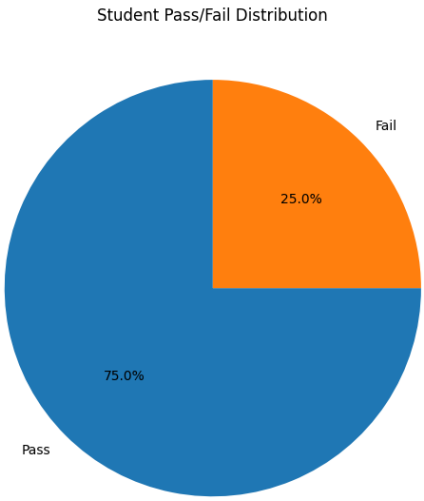
    Subject Name: Mathematics
    Average Marks: 71.45
    Highest Marks: 95
    Lowest Marks: 24
    Students Passed: 19
    Students Failed: 1

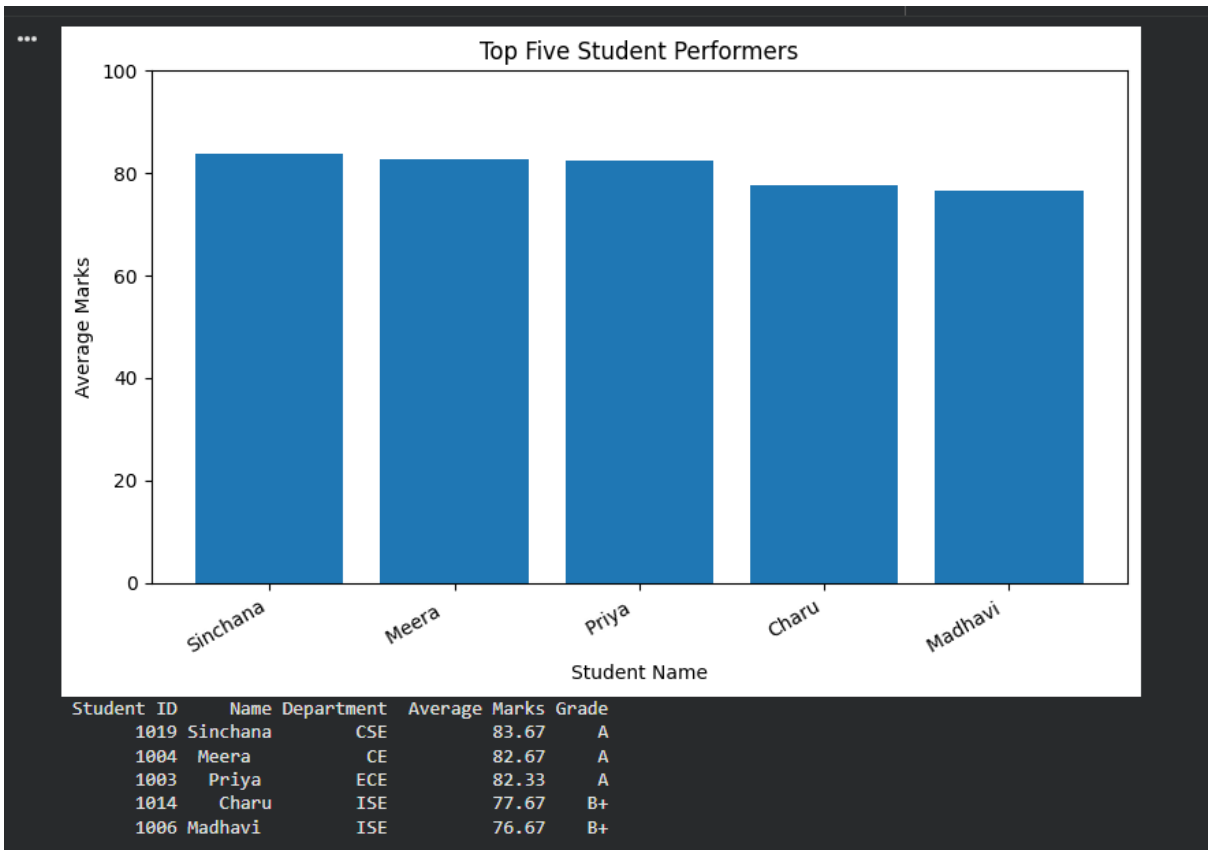
    Subject Name: Python programming
    Average Marks: 67.95
    Highest Marks: 99
    Lowest Marks: 32
    Students Passed: 18
    Students Failed: 2

    Subject Name: Engineering Drawing
    Average Marks: 68.0
    Highest Marks: 90
    Lowest Marks: 30
    Students Passed: 17
    Students Failed: 3
```

10) Data Visualization







Student ID	Name	Department	Average Marks	Grade
1019	Sinchana	CSE	83.67	A
1004	Meera	CE	82.67	A
1003	Priya	ECE	82.33	A
1014	Charu	ISE	77.67	B+
1006	Madhavi	ISE	76.67	B+

Final Outcome

The Student Performance Analytics system is built in order to automate the process of analyzing student academic performance using Python.

Using the combination of Pandas for data handling, NumPy for numeric computations, and Matplotlib for visualization, the project demonstrates the process of converting the structured data about students into valuable information about their performance.

The finished product is expected to help the users to get acquainted with student performance, compare the results between subjects, see which students have passed or failed, and see the top performers.

Challenges & Learning

- Some of the column names in CSV file were throwing error during execution. So, I used strip () to make it proper.
- It was different to collect and get random 20 students' data.
- I learned the fundamentals of python, variables, data types, loops, conditional statements, and functions.
- I learned how to analyze a random dataset.

GitHub Repository

Link: https://github.com/amrutham004/Student_Performance_Analytics_System